2 **(a)** Obeys the equation $PV = nRT$

No intermolecular forces

(b) **(i)**

$$pV = \frac{1}{3}Nm\langle c^2 \rangle$$

$$(2.12x10^7)(1.84x10^{-2}) = \frac{1}{3}(3.20)\langle c^2 \rangle$$

$$\langle c^2 \rangle = 365700$$

r.m.s speed = 604.7 m s^{-1}

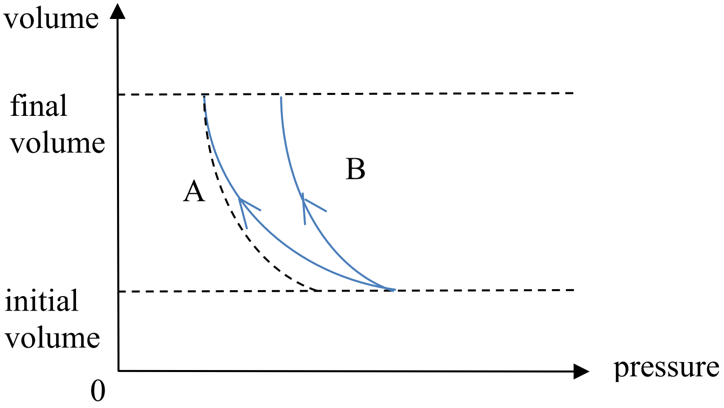
(ii) Heat supplied to system is zero

Work done on gas is negative

Change in internal energy is negative

So KE decreases

(iii)



1. Curve with correct direction

2. B is drawn correctly